SOMAIYA
VIDYAVIHAR
K J Somaiya Institute of Technology
(Formerly known as K J Somaiya Institute of Engineering and Information Technology)
An Autonomous Institute Permanently Affiliated to University of Mumbai.

T.Y. Mini Project Presentation

"Indian Meteorological Department Aviation Forecast Verification Web Portal"

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Introduction

- The project is developed under coordination with senior IMD officials, ensuring alignment with operational forecasting requirements.
- The web application has been hosted on IMD infrastructure and is currently in the testing and validation phase.
- The system is designed to support aviation weather forecasting and verification, which are critical for flight safety and decision-making.
- It covers three major sections:
  - Takeoff Forecast
  - Local Area Forecast
  - Aerodrome Warning Verification



Problem Statement

The current aviation forecast verification process at the Indian Meteorological Department (IMD) is largely manual, making it time-consuming, error-prone, and difficult to scale. The absence of an automated system for verifying Take-Off Forecasts (TOF), Local Area Forecasts (LAF), and Aerodrome Warnings (ADWR) delays performance evaluation and limits improvements in forecast accuracy and operational decision-making.

Objectives

- To automate the verification of aviation weather forecasts.
- To compare forecast data with observed METAR and upper-air observations.
- To verify Take-Off Forecasts, Local Area Forecasts, and Aerodrome Warnings.
- To evaluate forecast accuracy using standard meteorological verification criteria.
- To generate accuracy reports, graphs, and performance metrics.
- To support forecasters in improving the reliability and accuracy of aviation weather predictions

System Overview

Take-Off Forecast Verification

Purpose

- Automated verification of Take-Off forecasts using METAR observations
- User selects ICAO station & UTC period

Inputs

- Forecast: TAKE-OFF text files
- Observation: METAR reports

Verification Criteria

- Wind Direction: ±30°
- Wind Speed: ±5 knots
- Temperature: ±1°C
- QNH: ±1 hPa

Outputs

- Accuracy tables
- Daily accuracy graphs
- Forecast performance analysis

```
0000Z 120/05KT 28 1005 1006
0100Z 100/05KT 28 1006 1007
0200Z 110/05KT 29 1006 1007
0300Z 070/05KT 30 1007 1008
0400Z 140/05KT 31 1007 1008
0500Z 240/06KT 32 1007 1008
0600Z 260/08KT 32 1007 1008
0700Z 250/10KT 33 1006 1007
0800Z 260/10KT 33 1006 1007
0900Z 260/07G17KT 33 1005 1006
1000Z 270/07G17KT 32 1004 1005
1100Z 270/07G17KT 32 1004 1005
1200Z 270/10KT 31 1004 1005
```

Fig.1.1 Take-off forecast report

```
202509010000 METAR VABB 010000Z 26013KT 2500 BR SCT018 FEW025TCU BKN100 27/25 Q1004 NOSIG=
202509010030 METAR VABB 010030Z 26010KT 2500 BR SCT018 FEW025TCU BKN100 27/25 Q1005 NOSIG=
202509010100 METAR VABB 010100Z 26010G20KT 2500 -RA FEW012 SCT018 FEW025TCU BKN090 26/25 Q1005 NOSIG=
202509010130 METAR VABB 010130Z 24012KT 2500 BR FEW012 SCT018 FEW025TCU BKN090 27/25 Q1005 NOSIG=
202509010200 METAR VABB 010200Z 25012KT 2500 BR FEW012 SCT018 FEW025TCU BKN090 27/25 Q1005 BECMG 3000
202509010230 METAR VABB 010230Z 25014KT 3000 BR SCT018 FEW025TCU BKN100 28/25 Q1005 NOSIG=
202509010300 METAR VABB 010300Z 25014KT 3000 BR SCT018 FEW025TCU BKN100 28/25 Q1006 NOSIG=
202509010330 METAR VABB 010330Z 25010KT 3000 BR SCT018 FEW025TCU BKN100 28/25 Q1006 NOSIG=
202509010400 METAR VABB 010400Z 25013KT 3000 BR FEW012 SCT018 FEW030CB BKN090 28/25 Q1006 TEMPO 1500 S
202509010430 METAR VABB 010430Z 25012KT 3000 BR FEW012 SCT018 FEW025TCU BKN090 28/25 Q1006 NOSIG=
```

Fig.1.2 METAR observation report

VERIFICATION RESULT OF TAKE-OFF FORECAST
VABB
01/09/2025 00:00 UTC TO 30/09/2025 23:59 UTC

DAY	Wind Direction	Wind Speed	Temperature	QNH	Overall
01	95.8% (23)	75.0% (18)	95.8% (23)	100.0% (24)	70.8% (17)
02	87.5% (21)	91.7% (22)	95.8% (23)	100.0% (24)	83.3% (20)
03	91.7% (22)	100.0% (24)	95.8% (23)	100.0% (24)	87.5% (21)
04	61.9% (13)	76.2% (16)	61.9% (13)	100.0% (21)	38.1% (8)

Fig.1.3 Results of Take-off forecast verification

Local Area Forecast Verification

Purpose

- Automated verification of Local Area Forecasts using upper-air observations
- User-selected station & UTC period

Inputs

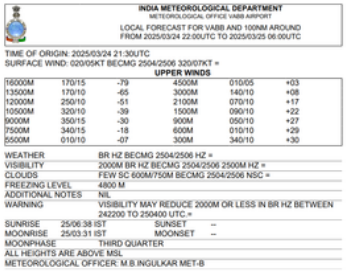
- Forecast: Local Area Forecast (PDF)
- Observation: Upper-air observations / CSV

Verification Criteria

- Temperature: ±2°C
- Wind Speed: ±10 knots
- Wind Direction: ±30°
- Weather: Occurrence / Non-occurrence

Outputs

- Parameter-wise accuracy
- Verification summary
- IMD-formatted reports



INDIA METEOROLOGICAL DEPARTMENT
METEOROLOGICAL OFFICE VABB AIRPORT
LOCAL FORECAST FOR VABB AND 100NM AROUND
FROM 2025/03/24 22:00UTC TO 2025/03/25 06:00UTC

TIME OF ORIGIN: 2025/03/24 21:30UTC
SURFACE WIND: 020/05KT BECMG 2504/2506 320/07KT =

UPPER WINDS

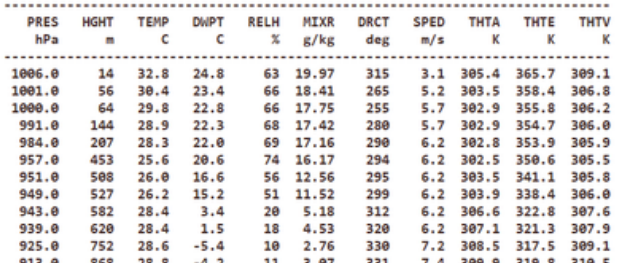
16000M	170/15	-79	4500M	010/05	+03
13500M	170/10	-65	3000M	140/10	+08
12000M	250/10	-51	2100M	070/10	+17
10500M	320/10	-39	1500M	090/10	+22
9000M	350/15	-30	900M	050/10	+27
7500M	340/15	-18	600M	010/10	+29
5500M	010/10	-07	300M	340/10	+30

WEATHER	BR HZ BECMG 2504/2506 HZ =
VISIBILITY	2000M BR HZ BECMG 2504/2506 2500M HZ =
CLOUDS	FEW SC 600M/750M BECMG 2504/2506 NSC =
FREEZING LEVEL	4800 M
ADDITIONAL NOTES	NIL
WARNING	VISIBILITY MAY REDUCE 2000M OR LESS IN BR HZ BETWEEN 242200 TO 250400 UTC =

SUNRISE	25/06:38 IST	SUNSET	--
MOONRISE	25/03:31 IST	MOONSET	--
MOONPHASE	THIRD QUARTER		

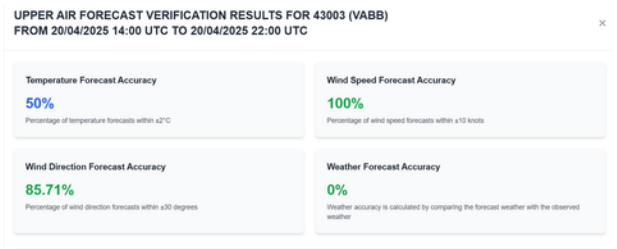
ALL HEIGHTS ARE ABOVE MSL
METEOROLOGICAL OFFICER: M.B.INGULKAR MET-B

Fig.2.1 Upper air forecast report



PRES hPa	HGHT m	TEMP C	DWPT C	RELH %	MIXR g/kg	DRCT deg	SPED m/s	THTA K	THTE K	THTV K
1006.0	14	32.8	24.8	63	19.97	315	3.1	305.4	365.7	309.1
1001.0	56	30.4	23.4	66	18.41	265	5.2	303.5	358.4	306.8
1000.0	64	29.8	22.8	66	17.75	255	5.7	302.9	355.8	306.2
991.0	144	28.9	22.3	68	17.42	280	5.7	302.9	354.7	306.0
984.0	207	28.3	22.0	69	17.16	290	6.2	302.8	353.9	305.9
957.0	453	25.6	20.6	74	16.17	294	6.2	302.5	350.6	305.5
951.0	508	26.0	16.6	56	12.56	295	6.2	303.5	341.1	305.8
949.0	527	26.2	15.2	51	11.52	299	6.2	303.9	338.4	306.0
943.0	582	28.4	3.4	20	5.18	312	6.2	306.6	322.8	307.6
939.0	620	28.4	1.5	18	4.53	320	6.2	307.1	321.3	307.9
925.0	752	28.6	-5.4	10	2.76	330	7.2	308.5	317.5	309.1

Fig.2.2 Upper air observation report



UPPER AIR FORECAST VERIFICATION RESULTS FOR 43003 (VABB)
FROM 20/04/2025 14:00 UTC TO 20/04/2025 22:00 UTC

Temperature Forecast Accuracy
50%
Percentage of temperature forecasts within ±2°C

Wind Speed Forecast Accuracy
100%
Percentage of wind speed forecasts within ±10 knots

Wind Direction Forecast Accuracy
85.71%
Percentage of wind direction forecasts within ±30 degrees

Weather Forecast Accuracy
0%
Weather accuracy is calculated by comparing the forecast weather with the observed weather

Fig.2.3 Results of Local Area forecast verification

Aerodrome Warning Verification

Purpose

- Automated verification of Aerodrome Warnings using METAR observations
- User-selected station & verification period

Inputs

- Forecast: Aerodrome Warning text files
- Observation: METAR reports

Verification Criteria

- Thunderstorm: Occurrence of TS, TSRA,etc.
- Gust/Wind: Wind speed & gusts exceed defined thresholds

Outputs

- Warning-wise accuracy
- Verification tables
- Accuracy summary cards
- Performance graphs

```
01
LIGHT AIRCRAFT WARNING FOR VABB - 20250901 00:30
-----------------------------------------
VABB 010000 AD WRNG 1 VALID 010030/010430
SFC WSPD 17KT MAX27 FROM WSW FCST NC=

LIGHT AIRCRAFT WARNING FOR VABB - 20250901 04:30
-----------------------------------------
VABB 010400Z AD WRNG 2 VALID 010430/010830
SFC WSPD 17KT MAX30 FROM W FCST NC=
```

Figure 3.1 WARNING INPUT DATA

```
'509010000 METAR VABB 010000Z 26013KT 2500 BR SCT018 FEW025TCU BKN100 27/25 Q1004 NOSIG=
'509010030 METAR VABB 010030Z 26010KT 2500 BR SCT018 FEW025TCU BKN100 27/25 Q1005 NOSIG=
'509010100 METAR VABB 010100Z 26010G20KT 2500  RA FEW012 SCT018 FEW025TCU BKN090 26/25 Q1005 I
'509010130 METAR VABB 010130Z 24012KT 2500 BR FEW012 SCT018 FEW025TCU BKN090 27/25 Q1005 NOSI
'509010200 METAR VABB 010200Z 25012KT 2500 BR FEW012 SCT018 FEW025TCU BKN090 27/25 Q1005 BECM
'509010230 METAR VABB 010230Z 25014KT 3000 BR SCT018 FEW025TCU BKN100 28/25 Q1005 NOSIG=
'509010300 METAR VABB 010300Z 25014KT 3000 BR SCT018 FEW025TCU BKN100 28/25 Q1006 NOSIG=
'509010330 METAR VABB 010330Z 25010KT 3000 BR SCT018 FEW025TCU BKN100 28/25 Q1006 NOSIG=
'509010400 METAR VABB 010400Z 25013KT 3000 BR FEW012 SCT018 FEW030CB BKN090 28/25 Q1006 TEMPO
'509010430 METAR VABB 010430Z 25012KT 3000 BR FEW012 SCT018 FEW025TCU BKN090 28/25 Q1006 NOSI
```

Figure 3.2 METAR DATA

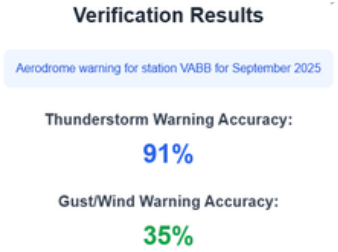


Figure 3.3 Results of Aerodrome warning

तत्त्व / Elements	विमान क्षेत्र चेतावनियों की सं. / Warning no 1 Warning no 2..... Warning no. (Issue date/Issue Time UTC)	रेंज वारे की प्रतिश occu
उष्णकटिबंधीय चक्रवात / Tropical cyclone	-	-
गर्जन सुनामी / Thunderstorms	04/0330,04/0730,13/0800,13/1230,13/1845,14/1600,14/2000,15/0000,15/0400,15/0800,16/0830,18/1030,18/1900,18/2300,19/0900,20/1645,21/0130,21/0800,21/1200,21/1600,21/2130,22/0130,22/0530,22/1400,23/0030,26/1100,27/0130,27/0530,27/0930,27/1330,27/1730,27/2130,28/0130,28/0530,28/0930,28/1330,28/2130	91%
ओला / Hail	-	-

Figure 3.4 Generated Report

AI-Based Aerodrome Warning Prediction

- **Logistic Regression:** Predicts Thunderstorm (TSRA) and Strong Wind/Gust (≥25 kt) events. Chosen because it is suitable for binary classification, computationally efficient, and provides interpretable probability scores.
- **Linear Regression:** Predicts surface wind speed, maximum gust, and wind direction. Chosen because it is simple, efficient, and suitable for estimating continuous weather parameters.

- **Input Features:** Historical METAR data, wind speed, gust, visibility, temperature, rainfall, and cloud indicators.
- **Prediction Process:** Uses current METAR observations to predict hazards 25–40 minutes ahead and generates alerts when validated probability thresholds are exceeded.

Model evaluation: The classifier achieved a nominal accuracy of 99.3% for thunderstorm prediction and 99.6% for strong-wind prediction on the chronological test set.

Block Diagram

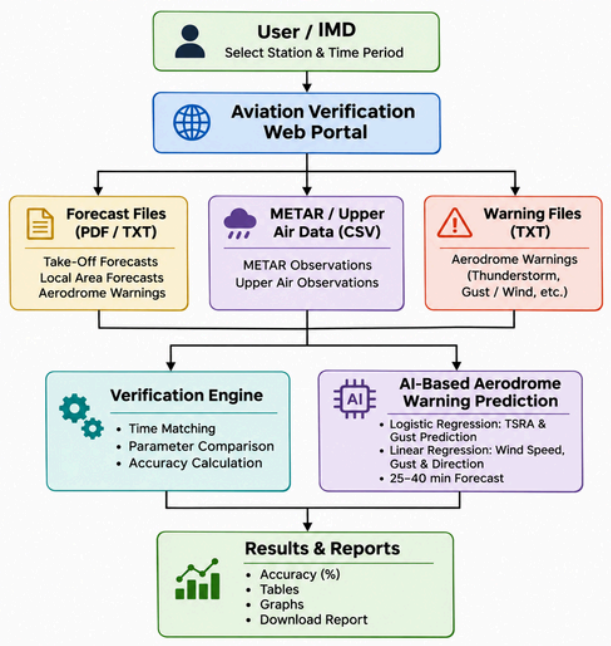


Figure 1: System Architecture of the Aviation Forecast Verification Web Portal

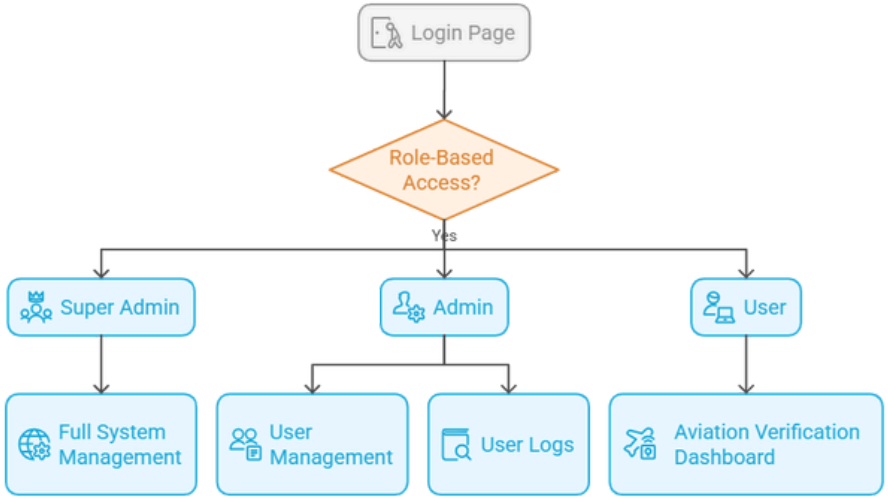


Figure 2: Role-Based Access Control (RBAC) Workflow

Results

- Successfully automated the verification of aviation weather forecasts, reducing manual effort and improving efficiency.
- Generated parameter-wise and overall forecast accuracy for Take-Off Forecasts, Local Area Forecasts, and Aerodrome Warnings.
- Compared forecast data with METAR and upper-air observations using standard IMD verification criteria.
- Produced detailed reports, accuracy tables, and graphical visualizations for comprehensive performance analysis.
- Predicts potential aerodrome weather warnings using historical METAR data and AI/ML models, providing warning details along with event probability for the selected forecast period.

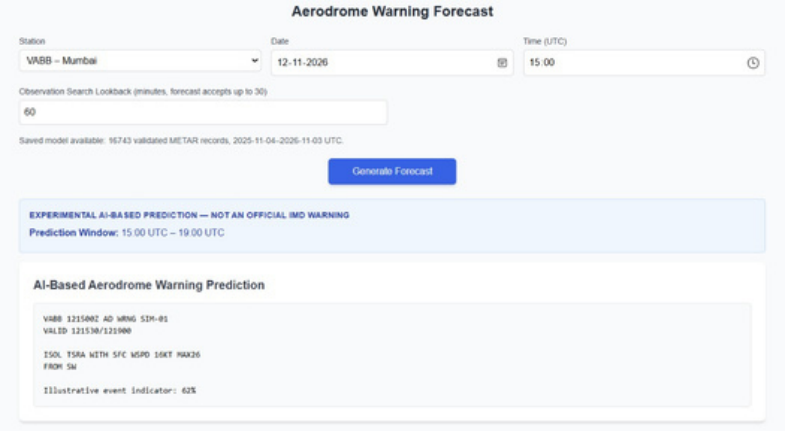


FIgure 4- AI-based prediction of potential aerodrome weather warnings using historical METAR data.

Applications

- Aviation weather forecast verification for airports.
- Performance evaluation of Take-Off and Local Area Forecasts.
- Verification of Aerodrome Warnings using observed METAR data.
- Decision support for aviation meteorologists and airport authorities.
- Forecast accuracy monitoring through reports and graphical analysis.
- Operational use at IMD for improving forecast quality and reliability.

Conclusion

A web-based system for automated aviation forecast verification was developed to streamline the verification of aviation weather forecasts using METAR and upper-air observations. The system improves the speed and accuracy of forecast evaluation, generates comprehensive reports and graphical analyses, reduces manual effort, and ensures consistent verification. This solution can assist the Indian Meteorological Department (IMD) in enhancing forecast quality, reliability, and overall operational efficiency.

References

- World Meteorological Organization (WMO) – Manual on Codes (WMO No. 306) - https://community.wmo.int/en/activity-areas/wis/publications/306-vI.1
- International Civil Aviation Organization (ICAO) – Annex 3: Meteorological Service for International Air Navigation - https://store.icao.int/en/annex-3-meteorological-service-for-international-air-navigation
- India Meteorological Department (IMD) – Official Website - https://mausam.imd.gov.in/
- OGIMET – METAR and Weather Data - https://www.ogimet.com/
- University of Wyoming – Upper Air Sounding Archive - https://weather.uwyo.edu/upperair/sounding.html
- World Meteorological Organization (WMO) – Forecast Verification Resources - https://community.wmo.int/en